Name:_____

One-sided Limits

Recall that $\lim_{x\to a} f(x) = L$ means that we can make $f(x)$ arbitrarily close to L by taking x sufficiently close to a.

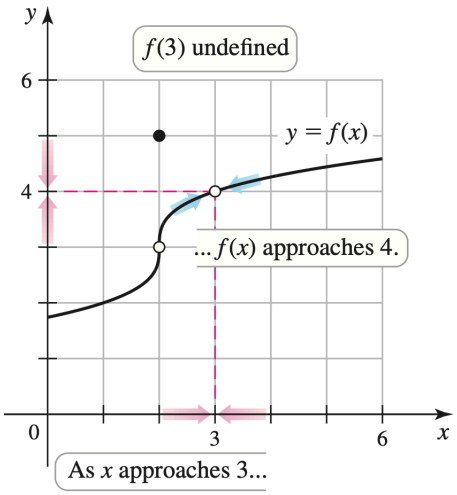


Figure 1: $\lim_{x\to 3} f(x) = 4$.

Note that this definition implies that $f(x)$ must approach L as x approaches a from both the left and from the right. In particular, if $f(x)$ approaches one value as x approaches a from the left and $f(x)$ approaches a different value as x approaches a from the right, then the limit does not exist. For example, consider the Heaviside function H, [1] defined by

$$H(t) = \begin{cases} 0, & \text{if } t < 0, \\ 1, & \text{if } t \geq 0. \end{cases}$$

The graph is

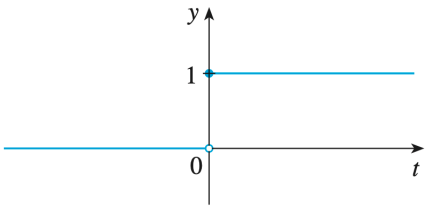


Figure 2: Graph of $H(t)$.

We see from the graph that $H(t)$ approaches 0 from the left, but approaches 1 from the right, so $\lim_{t\to 0} H(t)$ does not exist. However, the information about the function's behavior to the left or right is still often very useful, so we express this by writing

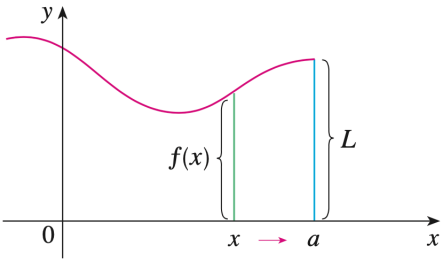
$$\lim_{t\to 0^-} H(t) = 0, \qquad \lim_{t\to 0^+} H(t) = 1.$$

We call these *one-sided limits*.

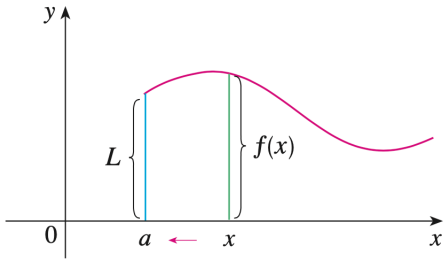
[1] Named after English mathematician and physicist Oliver Heaviside (1850-1925).

 Name:_____

The definitions of one-sided limits are similar to the definition of a limit: $\lim_{x\to a^-} f(x) = L$ means that we can make $f(x)$ arbitrarily close to L by taking x sufficiently close to a from the left. Similarly, $\lim_{x\to a^+} f(x) = L$ means that we can make $f(x)$ arbitrarily close to L by taking x sufficiently close to a from the right.



(a) $\lim_{x\to a^-} f(x) = L$



(b) $\lim_{x\to a^+} f(x) = L$

Their precise definitions are as follows:

Definition. (One-sided limits)

(a) Let f be defined on an open interval with a as its right endpoint, and let L be a real number. Then $\lim_{x\to a^-} f(x) = L$ means that given $\varepsilon > 0$ there exists a corresponding $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < a - x < \delta$.

(b) Let f be defined on an open interval with a as its left endpoint, and let L be a real number. Then $\lim_{x\to a^+} f(x) = L$ means that given $\varepsilon > 0$ there exists a corresponding $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < x - a < \delta$.

By comparing these definitions with the definition of a (two-sided) limit, we see the following:

Theorem. (Relationship between one-sided and two-sided limits) Let f be defined on an open interval containing a, except perhaps at a itself and let L be a real number. Then

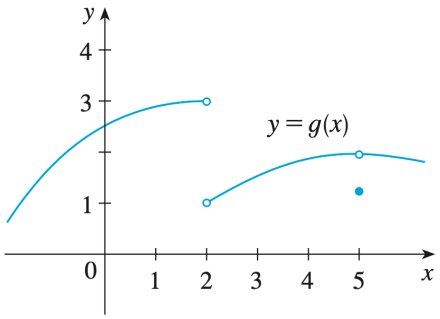
$$\lim_{x\to a} f(x) = L \text{ if and only if } \lim_{x\to a^-} f(x) = L \text{ and } \lim_{x\to a^+} f(x) = L.$$

That is, the (two-sided) limit exists and is equal to L if and only if both one-sided limits exist and are equal (to L).

Corollary. The Limit Laws all hold for one-sided limits.

Proof. We prove the Sum Law holds for right-sided limits; the proof for all other laws and cases are similar. If $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$, then by the theorem above $\lim_{x\to a^+} f(x) = L$ and $\lim_{x\to a^+} g(x) = M$. By the Sum Law for two-sided limits, $\lim_{x\to a}(f(x) + g(x)) = L + M$, so by the theorem above $\lim_{x\to a^+}(f(x) + g(x)) = L + M = \lim_{x\to a^+} f(x) + \lim_{x\to a^+} g(x)$. □

Example. Consider the function whose graph is shown below.



We see that $\lim_{x\to 2^-} g(x) = 3$ and $\lim_{x\to 2^+} g(x) = 1$, so $\lim_{x\to 2} g(x)$ does not exist. On the other hand, we have $\lim_{x\to 5^-} g(x) = 2 = \lim_{x\to 5^+} g(x)$, so $\lim_{x\to 5} g(x) = 2$.

Example. Consider again the Heaviside function

$$H(t) = \begin{cases} 0, & \text{if } t < 0, \\ 1, & \text{if } t \geq 0. \end{cases}$$

We now prove that $\lim_{t\to 0^-} H(t) = 0$ and $\lim_{t\to 0^+} H(t) = 1$. Let $\varepsilon > 0$. If $t < 0$, then $|H(t) - 0| = |H(t)| = |0| = 0 < \varepsilon$ automatically, so we can take any positive value of δ. This proves that $\lim_{t\to 0^-} H(t) = 0$. For $t \geq 0$, $|H(t) - 1| = |1 - 1| = |0| = 0 < \varepsilon$ automatically, so we can take any positive value of δ. This proves that $\lim_{t\to 0^+} H(t) = 1$.

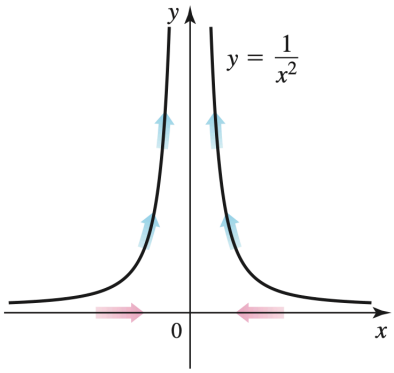
Name:_____

Infinite Limits

Consider now the graph of $f(x) = \frac{1}{x^2}$.

Table 2.5

x	$f(x) = 1/x^2$
± 0.1	100
± 0.01	10,000
± 0.001	1,000,000
$\downarrow$	$\downarrow$
0	∞



As $x \to 0$ from either side, $f(x)$ grows larger and larger. Because $f(x)$ does not approach a real number as $x \to 0$, $\lim_{x\to 0} f(x)$ does not exist. Nevertheless, it is conventional to use limit notation and write $\lim_{x\to 0} f(x) = \infty$. This just indicates that $f(x)$ attains arbitrarily large positive values as $x \to 0$. We call this an *infinite limit*.

Similarly, when $f(x)$ takes arbitrarily large (in absolute value) negative values as $x \to a$ from both sides, we indicate this by writing $\lim_{x\to a} f(x) = -\infty$. (Again, this is just notation - the limit does not exist.)

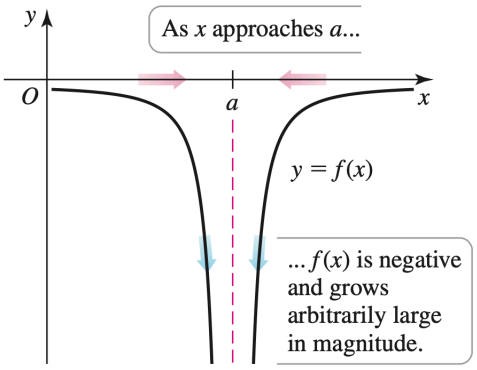
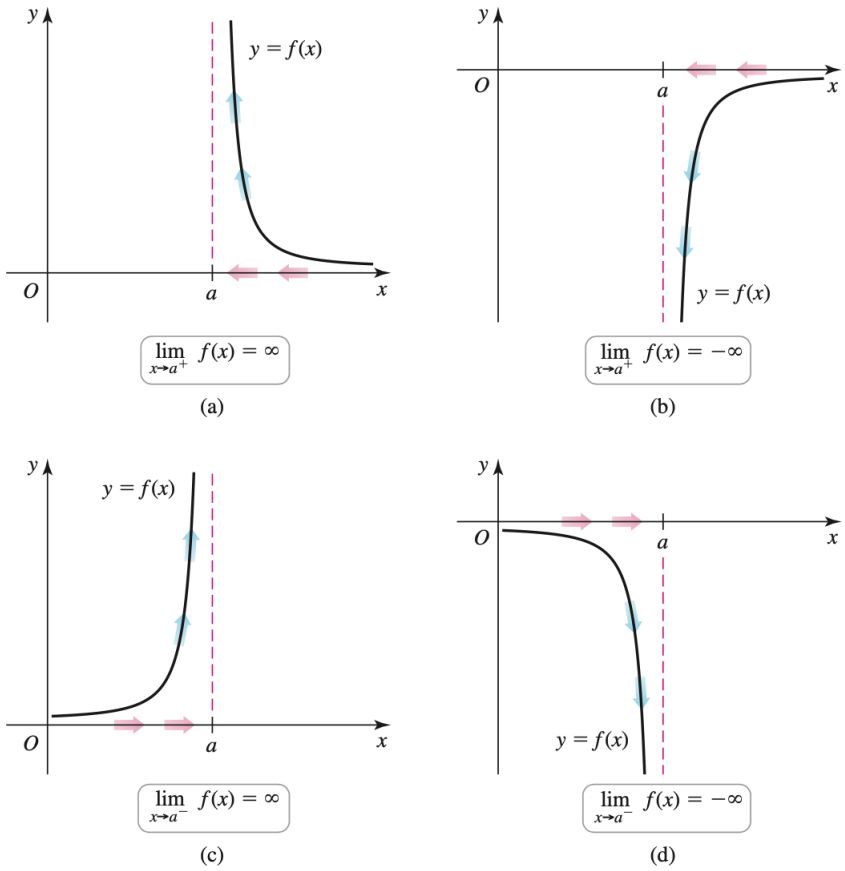


Figure 3: Graph of $f(x) = -\frac{1}{x^2}$.

Name:_____

And, of course, there are one-sided infinite limits:



In each example above, we see that the infinite limit arises from a vertical asymptote. In fact, we can take the existence of an infinite limit to be the *definition* of a vertical asymptote:

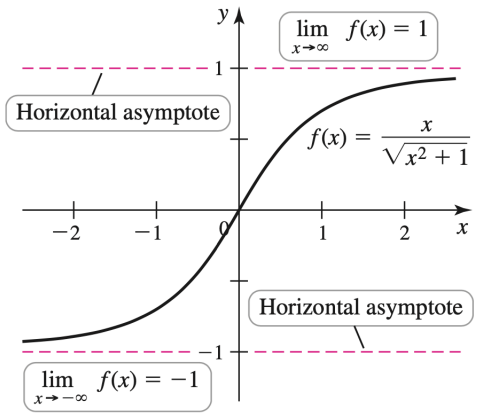
DEFINITION Vertical Asymptote

If $\lim_{x\to a} f(x) = \pm\infty$, $\lim_{x\to a^+} f(x) = \pm\infty$, or $\lim_{x\to a^-} f(x) = \pm\infty$, the line $x = a$ is called a **vertical asymptote** of f.

Name:_____

Limits at Infinity

Just as vertical asymptotes are associated with infinite limits, horizontal asymptotes are associated to *limits at infinity*. As you learned in previous courses, the question of the existence of a horizontal asymptote is a question about the *end behavior* of the function.



As before, we can use the language of limits at infinity to define horizontal asymptotes:

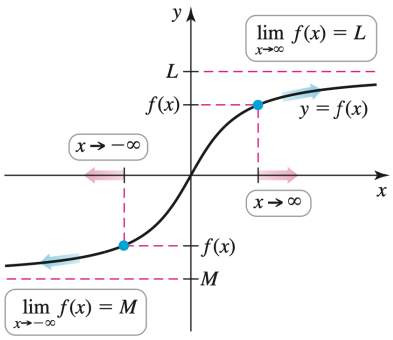


Figure 2.30

DEFINITION Limits at Infinity and Horizontal Asymptotes

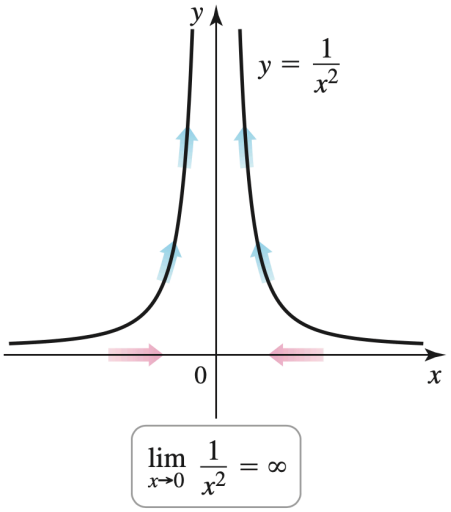
If $f(x)$ becomes arbitrarily close to a finite number L for all sufficiently large and positive x, then we write

$$\lim_{x\to\infty} f(x) = L.$$

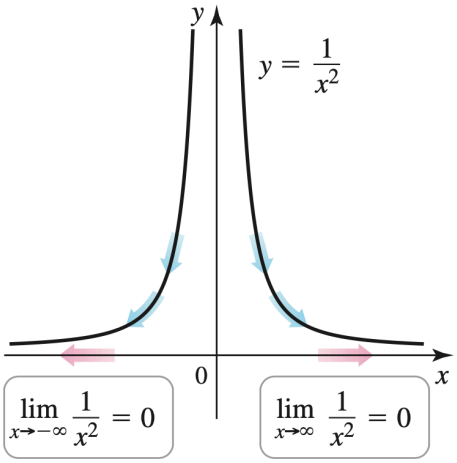
We say the limit of $f(x)$ as x approaches infinity is L. In this case, the line $y = L$ is a **horizontal asymptote** of f (Figure 2.30). The limit at negative infinity, $\lim_{x\to-\infty} f(x) = M$, is defined analogously. When this limit exists, $y = M$ is a horizontal asymptote.

Infinite Limits vs Limits at Infinity

Note the difference here between which variable (independent vs dependent) is approaching a real number and which one is approaching $\pm\infty$: for an infinite limit, x approaches a real number while $f(x) \to \pm\infty$, and correspondingly the graph has a vertical asymptote at $x = a$; for a limit at infinity, x approaches $\pm\infty$ while $f(x)$ approaches a real number, and correspondingly the graph has a horizontal asymptote.



(a) $f(x) \to \infty$ as $x \to a$



(b) $f(x) \to L$ as $x \to \pm\infty$

Infinite Limits at Infinity

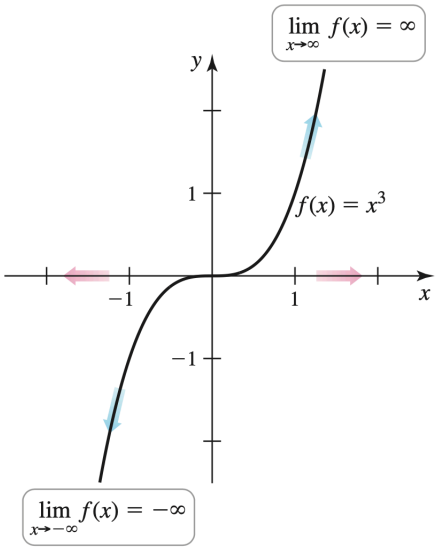
It is also possible to have an *infinite limit at infinity*:

DEFINITION Infinite Limits at Infinity

If $f(x)$ becomes arbitrarily large as x becomes arbitrarily large, then we write

$$\lim_{x\to\infty} f(x) = \infty.$$

The limits $\lim_{x\to\infty} f(x) = -\infty$, $\lim_{x\to-\infty} f(x) = \infty$, and $\lim_{x\to-\infty} f(x) = -\infty$ are defined similarly.



Precise definitions of infinite limits and limits at infinity

We can give precise definitions to the infinite limits and limits at infinity introduced above. There are many cases, so we don't list them all, but we give a couple examples to give the basic idea. The reader can easily supply definitions for the remaining cases, as needed.

Definition. (Infinite limit) Let f be defined on an open interval containing a but perhaps not at a itself. Then $\lim_{x\to a} f(x) = \infty$ means that, for every $\varepsilon > 0$ there exists a corresponding $\delta > 0$ such that $f(x) > \varepsilon$ whenever $0 < |x - a| < \delta$.

That is, we can make $f(x)$ arbitrarily large (greater than any given $\varepsilon > 0$) by taking x sufficiently close (but not equal) to a.

Example. Let $f(x) = \frac{1}{x^2}$. We will prove that $\lim_{x\to 0} f(x) = \infty$. Let $\varepsilon > 0$ and choose $\delta = \frac{1}{\sqrt{\varepsilon}}$. Then $0 < |x| < \delta \implies 0 < \frac{1}{\delta} < \frac{1}{|x|} \implies \frac{1}{\delta^2} < \frac{1}{x^2}$ (since δ and x are both nonnegative) and therefore

$$\begin{aligned} |f(x)| &= \left|\frac{1}{x^2}\right| \\ &= \frac{1}{x^2} \\ &> \frac{1}{\delta^2} \\ &= \frac{1}{(\frac{1}{\sqrt{\varepsilon}})^2} \\ &= \varepsilon, \end{aligned}$$

proving that $\lim_{x\to 0} f(x) = \infty$.

Definition. (Limit at infinity) Let f be defined on a semi-infinite interval of the form (a,∞) for some real number a, and let L be a real number. Then $\lim_{x\to\infty} f(x) = L$ means that, given $\varepsilon > 0$ there exists a corresponding $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $x > \delta$.

That is, we can make $f(x)$ arbitrarily close to L by taking x sufficiently large.

Example. Let $f(x) = \frac{1}{x}$. We will prove that $\lim_{x\to\infty} f(x) = 0$. Let $\varepsilon > 0$. Choose $\delta = \frac{1}{\varepsilon} > 0$. Then $x > \delta \implies x > \frac{1}{\varepsilon} \implies \varepsilon > \frac{1}{x} = \left|\frac{1}{x}\right|$ (since $x > 0$). This proves that $\lim_{x\to\infty} f(x) = 0$.

Limit Laws for infinite limits, limits at infinity

Theorem. (Limit laws for limits at infinity) The limit laws all hold for *finite* limits at infinity.

Proof. There are many cases to check. We prove that if $\lim_{a\to\infty} f(x) = L$ and $\lim_{a\to\infty} g(x) = M$, then $\lim_{a\to\infty}(f(x)+g(x)) = L+M = \lim_{a\to\infty} f(x) + \lim_{a\to\infty} f(x)$, and leave the others as exercises.

Let $\varepsilon > 0$. Since $\lim_{a\to\infty} f(x) = L$, there exists a corresponding $\delta_1 > 0$ such that $x > \delta_1$ implies $|f(x) - L| < \frac{\varepsilon}{2}$. Similarly, since $\lim_{a\to\infty} f(x) = L$, there exists a corresponding $\delta_1 > 0$ such that $x > \delta_2$ implies $|g(x) - M| < \frac{\varepsilon}{2}$. Take $\delta = \max\{\delta_1, \delta_2\}$. Then $x > \delta$ implies

$$\begin{aligned} |f(x) + g(x) - (L+M)| &= |f(x) - L + g(x) - M| \\ &\le |f(x) - L| + |g(x) - M| \quad \text{(by the Triangle Inequality)} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \quad \text{(since } x > \delta_1 \text{ and } \delta_2\text{)} \\ &= \varepsilon. \end{aligned}$$

□

Name:_____

The limit laws continue to hold when one or both of the limits is infinite, with some exceptions. To remember these rules, it helps to work in the *extended real number system*, which is defined to be the set $\mathbb{R} \cup \{\pm\infty\}$, where the two symbols $\pm\infty$ are defined to satisfy the following rules:

$$\begin{aligned}
x < \infty \quad & \text{for all } x \in \mathbb{R}, \\
-\infty < x \quad & \text{for all } x \in \mathbb{R}, \\
x + \infty = \infty + x = \infty \; & \text{for all } x \in \mathbb{R}, \\
x - \infty = -\infty + x = -\infty \; & \text{for all } x \in \mathbb{R}, \\
x \cdot \infty = \infty \cdot x = \begin{cases} \infty & \text{if } x > 0, \\ -\infty & \text{if } x < 0. \end{cases} & \\
x \cdot -\infty = -\infty \cdot x = \begin{cases} -\infty & \text{if } x > 0, \\ \infty & \text{if } x < 0. \end{cases} & \\
\frac{x}{\pm\infty} = 0 \text{ for all } x \in \mathbb{R}. &
\end{aligned}$$

Example.
We will prove that if $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = \infty$, then $\lim_{x\to a}(f(x)+g(x)) = \infty$.
Proof. Let $\varepsilon > 0$. Since $\lim_{x\to a} f(x) = L$ there exists $\delta_1 > 0$ such that $0 < |x-a| < \delta_1$ implies

$$\begin{aligned}
|f(x) - L| < \varepsilon &\iff -\varepsilon < f(x) - L < \varepsilon \\
&\iff L - \varepsilon < f(x) < L + \varepsilon.
\end{aligned}$$

Since $\lim_{x\to a} g(x) = \infty$, there exists $\delta_2 > 0$ such that $0 < |x-a| < \delta_2$ implies $g(x) > 2\varepsilon - L$. [2]
Taking $\delta = \min\{\delta_1, \delta_2\}$, it follows that $0 < |x-a| < \delta$ implies

$$f(x) + g(x) > L - \varepsilon + 2\varepsilon - L = \varepsilon.$$

In this sense, we have proved that if $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = \infty$, then

$$\lim_{x\to a}(f(x)+g(x)) = \infty = L + \infty = \lim_{x\to a} f(x) + \lim_{x\to a} g(x).$$

□

The following operations, on the other hand, are all left undefined:

$$\frac{0}{0}, \frac{\pm\infty}{\pm\infty}, 0 \cdot \pm\infty, \infty - \infty.$$

This is because none of these expressions can be defined unambiguously. For example, one defines $\frac{0}{0} = x$ for some real number x if and only if $0 = 0 \cdot x$, but this is true automatically for every real number x. Similarly, $\infty - \infty = x$ if and only if $\infty + x = \infty$, but this holds for every real number x.

These undefined expressions are often said to be *indeterminate forms* since when the right-hand-side of the limit laws equals one of these, the limit law cannot be used to determine the limit,

[2] One sees that $2\varepsilon - L$ is not necessarily positive. However, if one can make $g(x)$ greater than any positive number, then it follows immediately that one can make $g(x)$ greater than any number.

which may or may not exist.

Remark. It is a common misconception that, when a limit law produces an indeterminate form, then the limit doesn't exist. Instead, the correct conclusions is that the particular limit law used tells us *nothing* about the limit, and that we must try another technique to find the limit.

Example. Since $\lim_{x\to 0} x^n = 0$ for every positive integer n, the expressions $\frac{\lim_{x\to 0} x^3}{\lim_{x\to 0} x}$ and $\frac{\lim_{x\to 0} x}{\lim_{x\to 0} x^3}$ both result in the indeterminate form $\frac{0}{0}$. However,

$$\lim_{x\to 0} \frac{x^3}{x} = \lim_{x\to 0} x^2 = 0,$$

while

$$\lim_{x\to 0} \frac{x}{x^3} = \lim_{x\to 0} \frac{1}{x^2} = \infty.$$

This illustrates that an indeterminate form tells us nothing about the existence of a limit.